

Prithish Shan

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EDUCATION

Carnegie Mellon University

B.S. Statistics & Machine Learning, Computer Science

GPA: 3.66/4.00; Dean's List

Relevant Coursework: Principles of Imperative Computation, Principles of Functional Programming, Concepts of Mathematics (discrete), Matrices & Linear Transformations, Probability & Statistics, Computer Systems

Pittsburgh, PA

Aug 2025 – May 2029

WORK EXPERIENCE

Cofounder & Software Engineer | MyoVerse

Jan 2026 – present

- Pitched wireless surface electromyography (sEMG) platform for sports Physical Therapy to Mark Cuban and others; obtained \$300,000 in grants and credits from AWS, UPMC and CMU
- Designed, deployed, and continue to maintain software infrastructure from the ground up including:
 - C device firmware on Zephyr RTOS platform for robust wireless timekeeping and transmission of ultrahigh frequency EMG data packets
 - Flutter mobile application for seamless Bluetooth connection and assignment of sensors onto respective muscles, with controller-maintained state for reactivity, isolated thread workers for real-time digital signal processing (bandpass filtering and Fourier analysis), OAuth2 identity management, real-time cloud storage for data and long-term insights
 - TypeScript API service deployed on AWS lambda to store 2 million plus sensor logs in real time and serve long-term analytics to web and mobile consumers
 - Next.js [landing page](#) and data analytics platform for users to export session data and monitor recovery through biomarker trends

Research Intern | Mayo Clinic – Van Blitterswijk Lab

June 2023 – Apr 2024

- Developed machine learning pipeline in research setting to characterize novel RNA splicing in genetic cases of ALS and FTLD from long-read mRNA sequencing data
- Automated high-performance computing workflows with shell scripts, trained and optimized multilayer perceptron, support vector machine, and random forest models using Bayesian hyperparameter tuning and cross-validation
- Pipeline enabled us to identify prognostic RNA transcripts and train models for predicting clinical outcomes from unseen patient transcriptomic data
- Presented [research poster](#) to clinicians and researchers at Mayo Clinic, regional, and state conferences

SELECTED PROJECTS & AWARDS

Dermal (Acne Analysis) AI App

Oct 2025 – present

- Built and deployed production iOS (Swift) app with FastAPI backend that segments acne lesions from user photos using U-Net model trained on 1,200 dermatologist annotated images using Azure ML, and applies optical density physics to map hemoglobin and melanin distribution for lesion-type classification and quantification of abnormal pigmentation
- Launched to Apple App Store, gaining 100+ organic users and \$5,000 in funding through CMU's Swartz Center for entrepreneurship and AWS Startups.

Agentic Real-Estate Feasibility Platform

Jan 2026 – Mar 2026

- Architected and shipped MVP real estate feasibility platform using RAG for land discovery and analysis
- Designed geospatial data pipeline ingesting 100k+ land parcels into postGIS and indexing zoning documents with pgvector embeddings, and implemented scalable Express.js API integrating LangChain agents to power Angular frontend workflows
- Performed customer discovery with commercial real estate developers, including CEO of RISE Commercial

SAP Intelligence Track 1st Prize | ProdHacks @ CMU

Feb 2026

- Built generative AI ideation platform in 24 hours to allow product managers to generate and iterate on 3d models from just natural language input, awarded 1st prize by decision makers from SAP

Distributed Web GPU Computing Platform | TartanHacks

Feb 2026

- Built solution to remedy expensive GPU computing that splits parallelizable GPU tasks into jobs that clients can complete asynchronously from nothing but a browser tab
- Implemented Next.js app, Postgres database with concurrency handling, and centralized document storage in S3 in 24 hours

Congressional App Challenge | Office of Congressman Aaron Bean

Oct 2023 – Mar 2024

- Developed React Native mobile app and Node.js API to teach users about nearby historical sites via GPS and OpenAI API
- Won Congressional App Challenge representing FL-04 district and presented App at U.S. Capitol

SKILLS

AI/ML – PyTorch, Pandas, Langchain, Deep learning, RAG

Web & Mobile – Next.js, Angular, Flutter, Swift

API/CLOUD – Node, Docker, AWS (S3, RDS, Lambda)

Languages – C, Python, Java, TypeScript, Postgres SQL